FarmVerse – Precision Agriculture Management System

Requirement Analysis & System Design Document

1. Project Name

FarmVerse – Precision Agriculture Management System

2. Problem Statement

Existing System

Many farmers still manage farm information manually using notebooks or memory. This creates several challenges:

- Crop records may be lost.
- Irrigation schedules are difficult to track.
- Fertilizer usage is not recorded properly.
- Harvest history is unavailable.
- Farmers cannot easily monitor multiple farms.
- No centralized platform exists for farm management.

Proposed System

FarmVerse is a web-based Precision Agriculture Management System that helps farmers manage farm operations digitally. It allows users to maintain crop information, irrigation schedules, fertilizer records, and harvest history in one centralized platform.

3. Solution Architecture

```
Farmer
  │
  ▼
Frontend (React)
  │
HTTP REST API Requests
  │
  ▼
Backend (Spring Boot)
  │
CRUD Operations
```

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▼
MySQL Database

4. User Roles

Farmer

- Register
- Login
- Add Farm
- View Farm
- Update Farm
- Delete Farm
- Add Crop
- Update Crop
- Delete Crop
- View Reports

Admin

- Manage Farmers
- View All Farms
- Delete Invalid Records
- Generate Reports

5. Proposed Database Structure

Farmer

Fields
Farmer ID
Name
Age
Mobile Number
Email
Password
Village

Farm

Fields
Farm ID
Farmer ID
Farm Name

Farm Type
Location
Soil Type
Farm Area

Crop

Fields
Crop ID
Farm ID
Crop Name
Season
Planting Date
Expected Harvest
Crop Status

Irrigation

Fields
Irrigation ID
Farm ID
Irrigation Date
Water Quantity
Remarks

Fertilizer

Fields
Fertilizer ID
Farm ID
Fertilizer Name
Quantity
Application Date

6. Backend Modules (Spring Boot)

Farmer Module

- Create Farmer
- View Farmer
- Update Farmer
- Delete Farmer

Farm Module

- Add Farm

- View Farm
- Update Farm
- Delete Farm

Crop Module

- Add Crop
- View Crop
- Update Crop
- Delete Crop

Irrigation Module

- Schedule Irrigation
- View Schedule
- Update Schedule
- Delete Schedule

Fertilizer Module

- Add Fertilizer
- Update Fertilizer
- Delete Fertilizer
- View Fertilizer History

7. REST APIs

Farmer

Method	Endpoint
POST	/farmers
GET	/farmers
GET	/farmers/{id}
PUT	/farmers/{id}
DELETE	/farmers/{id}

Farm

Method	Endpoint
POST	/farms
GET	/farms
PUT	/farms/{id}
DELETE	/farms/{id}

Crop

Method	Endpoint

POST	/crops
GET	/crops
GET	/crops/{id}
PUT	/crops/{id}
DELETE	/crops/{id}

8. Frontend Pages (UI)

Public Pages

- Home
- About
- Contact
- Login
- Register

Farmer Dashboard

- Dashboard
- My Farms
- Crop Management
- Irrigation
- Fertilizer
- Reports
- Profile

Admin Dashboard

- Dashboard
- Farmer Management
- Farm Management
- Reports
- Settings

9. Functional Requirements

- User Registration
- User Login
- Manage Farms
- Manage Crops
- Schedule Irrigation
- Record Fertilizer Usage
- Generate Reports
- Profile Management

10. Non-Functional Requirements

- Secure Authentication
- Responsive User Interface
- Fast Performance
- Easy Navigation
- Reliable Database
- Data Validation
- Scalability

11. Technology Stack

Frontend

- React
- HTML
- CSS
- Bootstrap

Backend

- Java
- Spring Boot
- Spring MVC
- Spring Data JPA

Database

- MySQL

Tools

- Git
- GitHub
- Maven
- Postman
- IntelliJ IDEA